

Using the Linux Command Line Interface



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Overview



Becoming the CLI Guru

- Command line shortcuts
- Create directories
- Create files
- Redirecting output
- Basic file operations
- Aliases
- Compare files
- Edit files using nano and vim
- Edit files using sed



Shortcuts



Shell Shortcuts

Move to the start of the line **CTRL + a**

Move to the end of the line **CTRL + e**

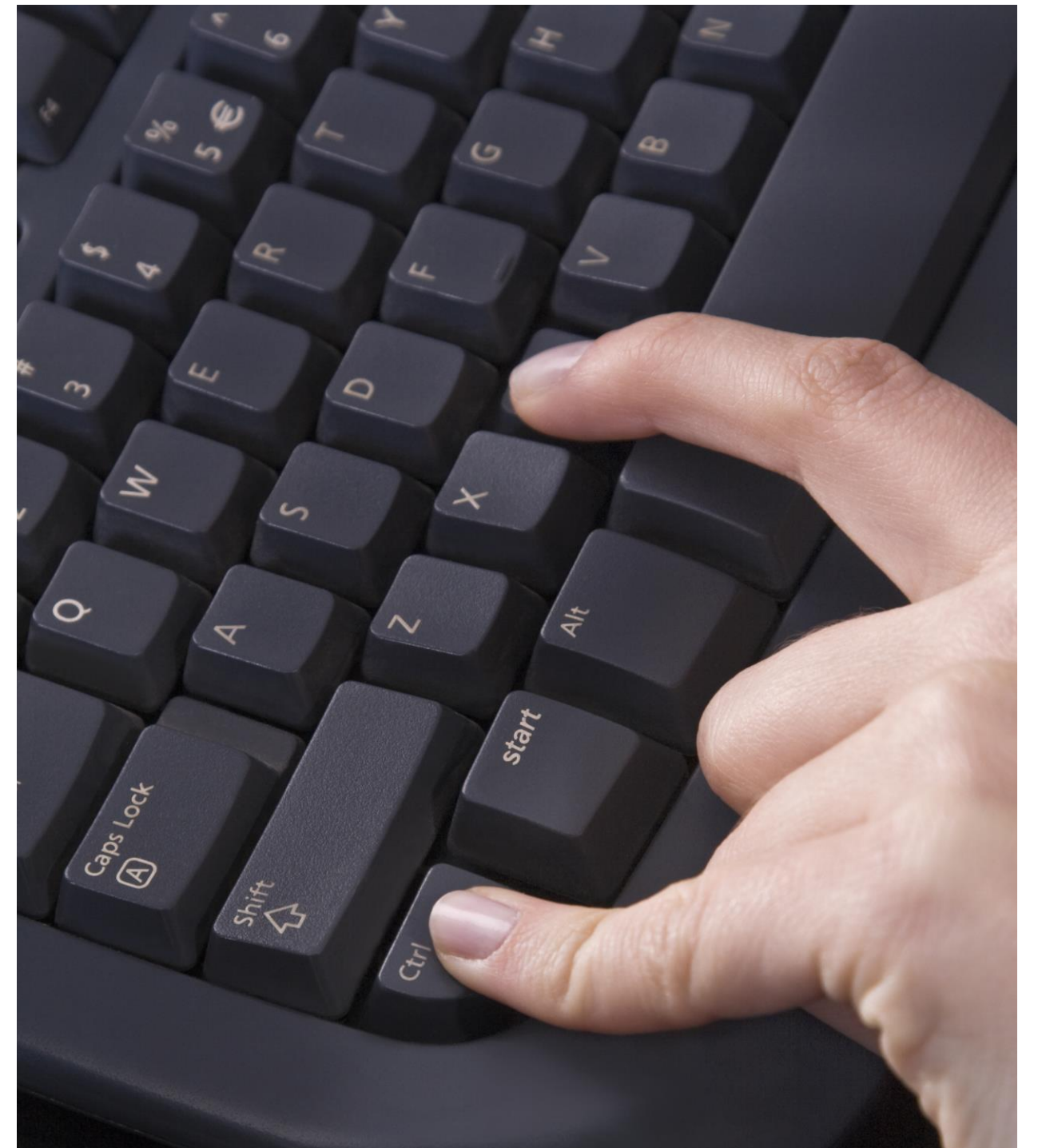
Repeat the last command and arguments !!

Repeat the last argument **ESC + .**

Move to your home directory **cd**

Move to previous directory **cd -**

Recall previous command **CTRL + r**



Demo



To begin, we use some of these shortcuts:

- Working on Ubuntu1
- Mastering shell shortcuts



Create Directories



```
$ mkdir ~/dir1
$ mkdir -p ~/dir2/dir3
$ mkdir -p ~/ubuntu/{14,16,18,20,22}.04/{gold,current}
$ sudo apt install -y tree
$ tree ubuntu
```

Creating Directories

The **mkdir** command is used to create directories. The option **-p** to create the parent directories. Using brace expansion to create complex directory structures

Demo



Let's create ourselves some directories



Create Files



```
$ touch file1  
$ ls -l file1  
$ file file1  
$ cat file1  
$ touch file1  
$ ls -l file1
```

Creating Files

We can create new empty files using the touch command. If the file already exists, we will update the last modified time stamp

Demo



Let's create files using touch



Redirection



```
$ > file2
$ ls -l /etc/hosts > out.txt
$ ls -l /etc/Hosts 2> err.txt
$ ls /etc/hosts /etc/Hosts &> combined.txt
$ ls -l | wc -l
```

Redirection

Output from a command can be sent to the file. To overwrite we use **>** and to append we use **>>**. The default is redirecting **STDOUT** but we can use 2 to redirect error output. The pipe or vertical bar takes the output of one command to the input of another.


```
$ echo "8.8.8.8 google" | sudo tee -a /etc/hosts
```

Tee

Using the command `tee`, we can send output to the screen and too a file. This is also useful where we need to raise our privileges, standard redirection works with the rights of the user that opened the shell.

Demo



Working with Redirection:

- > >>
- 1 / 2
- tee



File Operations



File Operations

cp

Copy requires read to the source file and write to the target directory

mv

With mv we can move or rename files and requires both read and write to the source and target directories

rm

The delete a file we can use rm, use rm -rf to delete a directory and its content



Demo



File Operations:

- cp
- mv
- rm



Aliases



Aliases

alias

Creates and displays aliases in your shell

unalias

Deletes aliases from your shell

Temporarily runs a command without its alias



Demo



Using Aliases:

- alias
- unalias
- \



Editing Files Using Nano



```
$ cp /etc/hosts .  
$ diff /etc/hosts hosts  
$ nano hosts  
$ diff /etc/hosts hosts
```

Editing and Comparing Files

The command `diff` can be used to display differences between two files. In Ubuntu, `nano`, is a simple text editor to get you started with text editing

Demo



Comparing files and using nano:

- diff
- nano



Editing Using Vim and Sed



```
$ vim +$ ~/hosts  
$ vim +/google/ ~/hosts  
$ vimtutor
```

Learning Vim

The text editor vim is very powerful, but that power does take a little time to harness. I will use vim throughout the course to help you learn more, but also use nano where you are more confident with that.

```
$ sed -i 's/google/googledns/' ~/hosts  
$ sed -i '$d' ~/hosts  
$ sed -i 'i1 '#This is a hosts file' ~/hosts
```

Programmatically Editing Files

Rather than interactively editing files, we can use sed to edit the files from the cli directly or within a script. We can edit existing lines or delete and add lines.

Demo



More Powerful Editing:

- Using vim
- Using sed



Summary



Using the Linux CLI

- Shortcuts
- Create directories and files
- Redirection, tee and pipe
- Copy, move and delete files
- Aliases
- Editing files



Up Next:
Managing Linux Permissions

